

Dabin Park

Ph.D. Student, School of Mathematics, Kyungpook National University

Daegu, Republic of Korea · Advisor: Prof. Sunghwan Moon

github.com/Dabin-Park

RESEARCH INTERESTS

Inverse problems; integral geometry and tomography (photoacoustic tomography, Radon-type transforms); operator learning and scientific machine learning for partial differential equations; finite element methods.

EDUCATION

Kyungpook National University — Ph.D. course, Mathematics 2023.09 – present

Kyungpook National University — M.S., Mathematics – 2023.08

- GPA 4.30 / 4.30

Kyungpook National University — B.S., Mathematics – 2022.02

- GPA 3.81 / 4.30; graduated with **highest honors** in the Department of Mathematics

PUBLICATIONS

Peer-reviewed journal articles

1. **D. Park**, S. Lee, S. Moon. *A machine learning and finite element framework for inverse elliptic PDEs via Dirichlet-to-Neumann mapping*. Journal of Computational and Applied Mathematics, **486** (2026), 117679. [doi:10.1016/j.cam.2026.117679](https://doi.org/10.1016/j.cam.2026.117679)
2. G. Hwang, G. Jeon, S. Moon, **D. Park**. *Implicit learning to determine variable sound speed and the reconstruction operator in photoacoustic tomography*. Applied Mathematics in Science and Engineering, **34**(1) (2026), 2661458. [doi:10.1080/27690911.2026.2661458](https://doi.org/10.1080/27690911.2026.2661458)
3. **D. Park**, S. Moon. *Reconstruction of an initial function from the solutions of the fractional wave equation on the light cone trace*. Mathematical Methods in the Applied Sciences, **47**(12) (2024), 9469–9475. [doi:10.1002/mma.10078](https://doi.org/10.1002/mma.10078)

In preparation

4. **D. Park**, S. Lee, S. Moon. *T-FEONet: A Transformer-based finite element operator network for time-dependent parametric PDEs*.

AWARDS

Best Paper Presentation Award, Fall Conference of the Korean Data & Information Science Society, Graduate Student Paper Presentation Competition 2023.11

Encouragement Prize, Annual Meeting of the Youngnam Mathematical Society 2023.07

Award for Presentation Excellence, 15th Graduate Student Workshop 2022.07

Highest Honors on B.S. graduation, Kyungpook National University 2022.02

CONFERENCE PRESENTATIONS

Implicit learning for the variable sound speed and the reconstruction operator in photoacoustic tomography

- **Oral.** Fall Conference of the Korean Data & Information Science Society, Daegu University, 2023.11.03–11.04. *Best Paper Presentation Award*

Reconstruction of the initial functions from the solutions of the fractional wave equation on the light cone trace

- **Poster.** Annual Meeting of the Youngnam Mathematical Society, Kyungpook National University, Daegu, 2023.07.14–07.16. *Encouragement Prize*
- **Oral (in English).** 15th Graduate Student Workshop, online, 2022.07.26–07.28. *Award for Presentation Excellence*
- **Oral.** Youngnam Mathematical Society, Dong-A University, Busan, 2022.06.23–06.24.
- **Poster.** KSIAM Spring Conference, Institute for Basic Science, 2022.05.27–05.29.
- **Oral.** Harmonic Analysis Workshop, Sungkyunkwan University, online, 2022.02.16–02.18.

CONFERENCES AND WORKSHOPS ATTENDED

ICIAM 2023, Waseda University, Tokyo, Japan	2023.08.20 – 08.25
KMS Spring Meeting, Daejeon Convention Center	2023.04.27 – 04.29
Winter AI Tutorial 2023 — Graph Neural Networks and Generative Models	2023.02.21 – 02.22
KSIAM Fall Conference, Sonocalm Jeju	2022.11.24 – 11.27
KSIAM Fall Conference, BEXCO, Busan	2021.12.02 – 12.05

COMPETITIONS

Nongnet Agricultural Price Fluctuation Forecasting AI Competition	2022.09
• Predicting the rate of change of agricultural prices against a reference date.	
• 18th place out of 69 teams. Competition page	

SKILLS

- **Scientific computing:** FEniCS / legacy dolfin, finite element methods
- **Machine learning:** PyTorch (CUDA), operator learning, Transformer architectures
- **Languages and tools:** Python, MATLAB, $\text{\LaTeX}$, Git, Docker / Dev Containers, Linux (WSL2)